Max Ellis *BDes (Hons) MSc AFHEA*

PhD Candidate in Psychoacoustics

Psychoacoustic Assessment and Machine Learning-Based Modelling of Novel Noise Source Perception in Complex Soundscapes

ellisacoustics.github.io · linkedin.com/in/psychoacoustic · github.com/ellisacoustics · orcid.org/0009-0007-5909-854X

Net Zero is bringing new technology into our towns and cities – **air taxis**, **electric cars**, **heat pumps** and **delivery drones** – with *sounds* we've never lived with before. I research the effect these *noise sources* have on us, *before* they arrive.

Noise is a "silent killer", second only to air pollution in its burden on health in Europe[WHO, 2011]. The next wave of technology will introduce noise that loudness alone cannot accurately judge[House of Lords, 2023]: the high-pitched whine of a rotor, the tonal hum of a compressor, rumbles of large aircraft flying low over gardens. My PhD research, sponsored by the Department for Environment, Food and Rural Affairs (Defra), seeks to accurately simulate these future soundscapes to gather response data from listening panels, for machine learning models to predict who will be annoyed, to what extent, and why. That gives regulators evidence to set standards on, and engineers a target for quieter machines, before they are everywhere.

Research Collaborations

Defra — PhD Sponsor
HEAD Acoustics GmbH — Journal Paper
Joby Aviation Ltd. — Journal Paper

Publications

2027

Multi-Model Analysis for Prediction of Perceived Annoyance from UAVs

M. W. Ellis, T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, Z. Podwińska, A. J. Torija
Journal of the Acoustical Society of America (JASA), 2027.

2027

Psychoacoustic Assessment of UAV Noise in Varying Ambiences, Operational Conditions and Contexts

M. W. Ellis, Z. Podwińska, A. J. Torija

Forum Acusticum 2027, Brussels.

Awarded Best Student Paper

2027

Impacts on Soundscape Perception from eVTOL Flyovers in Existing Urban Ambiences

M. W. Ellis, N. Shrubsole-Green, J. Bain, Z. Podwińska, A. J. Torija

AIAA 2027, San Diego.

2026

Comparing a General Sound Quality Metric for Aircraft Noise Using a Pre-Trained Neural Network Model to Traditional Psychoacoustics

T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, **M. W. Ellis**, A. J. Torija

Quiet Drones 2026, Delft. Human Response to UAS and UAM Noise. With HEAD Acoustics GmbH.

PDF

2025

Comparison of Deep Learning and Psychoacoustic Models to Predict UAV Noise Impact in Soundscapes

M. W. Ellis, M. C. Green, M. J. B. Lotinga, A. J. Torija

Forum Acusticum EuroNoise 2025, 11th Convention of the European Acoustics Association, Málaga, pp. 2121–2127.

doi:10.61782/fa.2025.0235 · PDF

All publications on Google Scholar · ORCID

Industry Experience

Aug 2027 – Oct 2027

Research Engineer (PhD Placement)

Dyson

- Assessing whether structural vibration can underpin sound quality metrics for product components, comparing vibration and sound quality data across anechoic, semi-reverberant and reverberant conditions.
- Applying transfer learning across sound quality metrics and jury test data from other products, to build machine learning models that predict listener response and satisfaction.

Aug 2026 – Oct 2026

Specialist Consultant (PhD Placement)

Noise Consultants Ltd. (part of Logika Group)

- Validated open-source noise measurements against modelled Defra noise-mapping data for infrastructure assessment.
- Ran a field campaign on an airport expansion project comparing physical and perceptual dominance of aircraft noise.

Academic Experience

Sep 2026 – May 2027

Visiting Lecturer in Music Technology

University of the West of Scotland

- 1st Supervisor on the Major Project (Dissertation) module for BSc (Hons) Music Technology students specialising in Acoustics and Environmental Noise.

2026 – Present

Associate Lecturer in Computer Science

St Mary’s University, Twickenham

- Teaching, marking and moderation across the online MSc Computer Science suite.
- Module reviewer and author.

2026 – Present

Teaching Assistant in Applied Acoustics

University of Salford

- Lab demonstrator for Environmental Noise Measurement and Modelling (BEng/MSc).
- Tutor for Immersive Sound Reproduction (MSc/PgDip).

Education

2025 – 2028

PhD Acoustics and Audio Engineering

University of Salford Acoustics Innovation Institute

Year 2

2023 – 2024

MSc Computer Science with Artificial Intelligence

Abertay University

Distinction

2021 – 2023

BDes (Hons) Sound for Moving Image

University of Glasgow

2nd Class Honours, Upper-Division (2:1)

2019 – 2021

DipHE Music Technology

University of the West of Scotland

Distinction

🎖 Awarded BSc Level 8 Court Medal

2019

BTEC Level 3 Digital Synthesis and Sampling

Fife College

Pass

2018 – 2019

SCQF Level 6 Sound Production

Glasgow Clyde College Anniesland

Pass

Working Groups and Committees

Institute of Acoustics Scottish Branch Committee — Member; Early Careers Group — Member

NASA Unmanned Aircraft Systems Noise Working Group (UNWG) — Member

Noise Network Plus A.I. and Digital Working Group — Member

Software Proficiencies

Max/MSP Ableton ArtemiS SUITE R Python QGIS LaTeX (Overleaf) REAPER MATLAB TensorFlow PyTorch Feature Manipulation Engine (FME)

Certifications

Advanced Prompt Engineering
Vanderbilt University

Show credential ↗

Advanced R Programming
Johns Hopkins University

AI for Noise Pollution Control
LSIB

Credential ID
2245-32007-113

Applications of AI in Acoustics
European Acoustics Association
Summer School

Auralization
Autumn School Series in Acoustics (ASSA) 2026

Digital Signal Processing
EPFL

Show credential ↗

Ethics of Artificial Intelligence
Politecnico di Milano

Show credential ↗

Responsible Research and Innovation
ORBITRRI

Transformer Models and BERT Model
Google

Trustworthy Generative AI
Vanderbilt University

Show credential ↗